11-2. Instrumental Variables

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Instrumental Variables* (Prof. Sanghack Lee, GSDS SNU)

The central question is how a causal effect can be learned when confounders are unmeasured and no adjustment set exists, by exploiting a variable that moves the treatment but affects the outcome only through it.

These notes consolidate the slides, derive the IV, ILS and 2SLS estimators and show their equality, develop the Angrist–Imbens–Rubin potential-outcome treatment of non-compliance with full identification of the complier effect, work the numerical CACE example, and summarize the practical issues (weak and many instruments). Physical PDF page numbers are listed in Section F.

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1 Motivation

Purpose. See instruments in action and state what makes an instrument good.

1.1 Instrumental variables in action

Study	Instrument Z	Treatment X	Outcome Y
Angrist and Krueger (1991)	quarter of birth	years of schooling	earnings
Card (1995)	proximity to college	years of schooling	earnings
Acemoglu et al. (2001)	settler mortality	institutional quality	GDP per capita
Miguel et al. (2004)	rainfall shocks	economic growth	civil conflict
Levitt (1996)	overcrowding litigation	prison population	crime rate

Returns to education. Ability U confounds schooling X and earnings Y. Angrist and Krueger: compulsory-schooling laws interact with quarter of birth; those born early in the year can drop out sooner and get less schooling. Card: growing up near a four-year college lowers the cost of attending, so proximity leads to more education and, through it, higher earnings.
Institutions and development. Geography and culture U confound institutions X and GDP Y. Acemoglu, Johnson and Robinson: high settler mortality led to extractive institutions, low mortality to inclusive ones; the institutional differences persist and explain modern income gaps. Miguel et al. use rainfall as an instrument for economic growth to study civil conflict in Africa.

1.2 Why instruments?

So far effects were identified by back-door adjustment, which requires measuring all confounders. If some are unmeasured, no adjustment set blocks the confounding. Idea: find Z that affects Y only through X ($Z \to X \to Y$, $X \leftarrow U \to Y$, $Z \perp\!\!\!\perp U$). Such Z is an *instrumental variable*.

1.3 Conditions for a good instrument

IVs are very popular in economics, but good ones are hard to find. A good instrument must satisfy

1. **Relevance:** a strong effect on the treatment (large $\mathrm{Cov}(X, Z)$); otherwise the IV estimate has large variance;
2. **Exclusion restriction:** no direct effect on the outcome;

and (implicitly) no confounding between Z and Y.

2 IV under linearity

Purpose. Derive the classical IV estimators in the linear constant-effect model.

2.1 The econometric setup

Consider $Y = \beta_0 + \beta_1 X + \beta_2 W + \varepsilon$ with scalar treatment X and covariates W; the estimand is β_1. X is *endogenous* (correlated with ε, i.e. confounded) and W is *exogenous* (uncorrelated with ε). Because ε depends on X, OLS of β_1 is biased. An instrument is a K-vector Z with $(Z, W) \perp \varepsilon$

(uncorrelatedness: $\mathrm{Cov}[\varepsilon, Z] = \mathrm{Cov}[\varepsilon, W] = 0$). If $K > 1$ the model is over-identified; if $K = 1$, just-identified.

2.2 Just-identified, no covariates

IV estimator. The ratio of covariances:

$$\hat{\beta}_{1,\mathrm{iv}} = \frac{\widehat{\mathrm{Cov}}(Y_i, Z_i)}{\widehat{\mathrm{Cov}}(X_i, Z_i)} = \frac{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})(Z_i - \bar{Z})}.$$

With a binary instrument this is the *Wald estimator*

$$\hat{\beta}_{1,\mathrm{iv}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0},$$

where $\bar{Y}_z, \bar{X}_z$ are sample means within group $Z = z \in \{0, 1\}$ (since $\widehat{\mathrm{Cov}}(Y, Z) = \hat{p}(1-\hat{p})(\bar{Y}_1 - \bar{Y}_0)$ with $\hat{p}$ the fraction with $Z = 1$, and likewise for X).

Indirect least squares (ILS). Fit two reduced-form regressions

$$Y = \pi_{10} + \pi_{11} Z + \varepsilon_1, \qquad X = \pi_{20} + \pi_{21} Z + \varepsilon_2;$$

the ratio of OLS slopes gives $\hat{\beta}_{1,\mathrm{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21} = \hat{\beta}_{1,\mathrm{iv}}$, since each slope is $\widehat{\mathrm{Cov}}(\cdot, Z)/\widehat{\mathrm{Var}}(Z)$.

Two-stage least squares (2SLS). Stage 1: regress X on Z, $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z$. Stage 2: regress $Y = \beta_0 + \beta_1 \hat{X} + \varepsilon$ by OLS to get $\hat{\beta}_{1,\mathrm{2sls}}$. Intuition: the instrument recovers the "unconfounded portion" of X. Since $\widehat{\mathrm{Cov}}(Y, \hat{X}) = \hat{\pi}_{21}\widehat{\mathrm{Cov}}(Y, Z)$ and $\widehat{\mathrm{Var}}(\hat{X}) = \hat{\pi}_{21}^2 \widehat{\mathrm{Var}}(Z)$, $\hat{\beta}_{1,\mathrm{2sls}} = \widehat{\mathrm{Cov}}(Y, Z)/(\hat{\pi}_{21}\widehat{\mathrm{Var}}(Z)) = \hat{\beta}_{1,\mathrm{iv}}$. Hence $\hat{\beta}_{1,\mathrm{iv}} = \hat{\beta}_{1,\mathrm{ils}} = \hat{\beta}_{1,\mathrm{2sls}}$.

2.3 Just-identified, with covariates

ILS: $Y = \pi_{10} + \pi_{11} Z + \pi_{12} W + \varepsilon_1$, $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $\hat{\beta}_{1,\mathrm{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21}$. 2SLS: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z + \hat{\pi}_{22} W$, then $Y = \beta_0 + \beta_1 \hat{X} + \beta_2 W + \varepsilon$; $\hat{\beta}_{1,\mathrm{2sls}}$ is the OLS coefficient of $\hat{X}$.

IV derivation. Given $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$ and $Y = \beta_0 + \beta_1 X + \underbrace{\beta_2 W + \varepsilon}_{\eta}$, assume in addition $\mathrm{Cov}[W, Z] = 0$. Then $\mathrm{Cov}[X, Z] \neq 0$ but $\mathrm{Cov}[\eta, Z] = 0$, so

$$\mathrm{Cov}[Y, Z] = \mathrm{Cov}[\beta_0 + \beta_1 X + \eta, Z] = \beta_1 \,\mathrm{Cov}[X, Z],$$

so

$$\beta_1 = \frac{\mathrm{Cov}[Y, Z]}{\mathrm{Cov}[X, Z]} = \frac{\mathrm{Cov}[Y, Z]/\,\mathrm{Var}[Z]}{\mathrm{Cov}[X, Z]/\,\mathrm{Var}[Z]} = \frac{\text{reduced form}}{\text{first stage}}.$$

2SLS derivation. Substituting X into the outcome model,

$$\begin{aligned} Y &= \beta_1(\pi_{22} W + \pi_{21} Z + \pi_{20} + \varepsilon_2) + \beta_2 W + \beta_0 + \varepsilon \\ &= \beta_1 \pi_{21} Z + (\beta_2 + \beta_1 \pi_{22}) W + \underbrace{\big(\beta_1(\pi_{20} + \varepsilon_2) + \beta_0 + \varepsilon\big)}_{\xi}, \end{aligned}$$

hence

$$Y = \beta_1(\pi_{22} W + \pi_{21} Z) + \beta_2 W + \xi = \beta_1 \underbrace{\mathbb{E}[X \mid W, Z]}_{\text{population analogue of } \hat{X}} + \beta_2 W + \xi,$$

where π_{20} is absorbed into ξ, so $\mathbb{E}[X \mid W, Z] - \pi_{20} = \pi_{21} Z + \pi_{22} W$ (intercept-free). Regressing Y on $\mathbb{E}[X \mid W, Z]$ and W recovers β_1; the sample analogue replaces it by the first-stage fitted value $\hat{X}$.

2.4 From linear to nonparametric IV

The linear analysis assumes a constant effect β_1 for everyone, and then IV = ILS = 2SLS. If effects are heterogeneous ($Y_i(1) - Y_i(0)$ varies), what does the IV estimand identify? A key setting is the randomized experiment with non-compliance, in which random assignment Z instruments treatment received X; Angrist, Imbens and Rubin (1996) give a potential-outcome framework.

3 Randomized experiments with non-compliance

Purpose. Identify the effect for compliers under random assignment, monotonicity and exclusion.

3.1 Setting

A drug is tested in an RCT: Z = random assignment, X = treatment actually received, Y = health outcome. Some patients assigned to treatment refuse the drug; some controls obtain it elsewhere. Comparing $X = 1$ with $X = 0$ is confounded (compliance is self-selected). Use Z as an instrument: it is randomized (independent of potential outcomes) and affects Y only through X (exclusion).

Potential outcomes (Angrist, Imbens and Rubin 1996, JASA). Outcomes are indexed by assignment: $Y(z)$, $z = 0, 1$. Treatment received is a post-assignment variable with potential values $X(z)$. Observed data: Z, $X = X(Z)$, $Y = Y(Z)$. Units are divided into latent *compliance types* $S = (X(0), X(1))$:

	$X(0) = 0$	$X(0) = 1$
$X(1) = 0$	never-taker ($X = 0$ always)	defier (X opposes Z)
$X(1) = 1$	complier (X follows Z)	always-taker ($X = 1$ always)

Only compliers change treatment when Z changes. S is latent, and each observed (Z, X) cell is a mixture: $(0,0)$: {C, NT}; $(0,1)$: {AT, D}; $(1,0)$: {NT, D}; $(1,1)$: {C, AT}. Additional assumptions are needed to identify type-specific effects. Goal: CACE = $\mathbb{E}[Y(1) - Y(0) \mid \text{complier}] = \text{ITT}_y/\text{ITT}_x$.

3.2 Intent-to-treat effects

S does not depend on Z; it is a baseline characteristic. Define type-specific ITT effects $\text{ITT}_s = \mathbb{E}[Y(1) - Y(0) \mid S = s]$, $s \in \{c, n, a, d\}$. With π_s the proportion of type s, the global ITT $\text{ITT}_y := \mathbb{E}[Y(1) - Y(0)]$ is

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \text{ITT}_n + \pi_a \text{ITT}_a + \pi_d \text{ITT}_d.$$

3.3 Identification assumptions

A1 (random assignment) $\{Y(0), Y(1), X(0), X(1)\} \perp\!\!\!\perp Z$.

A2 (strong monotonicity) (1) no defiers: $X(1) \geq X(0)$ for all units;

(2) compliers exist: $\mathbb{E}[X \mid Z = 1] \neq \mathbb{E}[X \mid Z = 0]$. Monotonicity often holds by design; with one-sided compliance $X(0) = 0$.

A3 (exclusion restriction for non-compliers) $Y(0) = Y(1)$ for all units with $S \in \{n, a\}$. For non-compliers X is the same under both assignments, yet Z could still affect Y directly

(e.g. a placebo effect); A3 rules out such direct $Z \to Y$ paths, the same "no direct effect" assumption as in linear IV, phrased in potential-outcome language.

3.4 Identification

A2 gives $\pi_d = 0$; A3 gives $\mathrm{ITT}_n = \mathrm{ITT}_a = 0$. Hence

$$\mathrm{ITT}_y = \pi_c \mathrm{ITT}_c + \pi_n \cdot 0 + \pi_a \cdot 0 + 0 \cdot \mathrm{ITT}_d = \pi_c \mathrm{ITT}_c, \qquad \mathrm{ITT}_c = \mathrm{ITT}_y / \pi_c.$$

With $0 < \pi_c < 1$, $|\mathrm{ITT}_y| < |\mathrm{ITT}_c|$: the global ITT is a conservative (attenuated) estimate.

Compliance-type proportions. Under A1 and A2, when $Z = 0$ only always-takers have $X = 1$, and when $Z = 1$ only never-takers have $X = 0$:

$$\pi_a = P(X = 1 \mid Z = 0) = \mathbb{E}[X \mid Z = 0], \qquad \pi_n = P(X = 0 \mid Z = 1) = 1 - \mathbb{E}[X \mid Z = 1],$$

$$\pi_c = 1 - \pi_a - \pi_n = \mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0].$$

Observed cells under monotonicity. $(Z, X) = (0, 0)$: compliers (with $X(0) = 0$) and never-takers; $(0, 1)$: always-takers only; $(1, 0)$: never-takers only; $(1, 1)$: compliers and always-takers. Two cells are pure and directly identify non-complier means; two are mixed, but the complier means can be solved for using the known proportions:

$$\mathbb{E}[Y \mid X = 0, Z = 0] = \tfrac{\pi_c}{\pi_c + \pi_n} \mathbb{E}[Y(0) \mid \text{complier}] + \tfrac{\pi_n}{\pi_c + \pi_n} \mathbb{E}[Y(0) \mid \text{never-taker}],$$
$$\mathbb{E}[Y \mid X = 0, Z = 1] = \mathbb{E}[Y(0) \mid \text{never-taker}], \qquad \mathbb{E}[Y \mid X = 1, Z = 0] = \mathbb{E}[Y(1) \mid \text{always-taker}],$$
$$\mathbb{E}[Y \mid X = 1, Z = 1] = \tfrac{\pi_c}{\pi_c + \pi_a} \mathbb{E}[Y(1) \mid \text{complier}] + \tfrac{\pi_a}{\pi_c + \pi_a} \mathbb{E}[Y(1) \mid \text{always-taker}].$$

(Here $Y(0)$, $Y(1)$ denote the outcomes under the treatment actually received, which for compliers coincide with the assignment-indexed ones.)

ITT on treatment. $\mathrm{ITT}_x = \mathbb{E}[X(1) - X(0)] = \pi_c$ by randomization: under monotonicity only compliers switch, so the fraction who switch is exactly π_c.

Theorem 3.1 (CACE / LATE; Imbens and Angrist 1994, Angrist et al. 1996)**.** *Under A1–A3,*

$$\mathrm{CACE} \equiv \mathbb{E}[Y(1) - Y(0) \mid S = c] = \frac{\mathrm{ITT}_y}{\mathrm{ITT}_x} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}.$$

The CACE (local average treatment effect, LATE) is a ratio of two ITT effects; it equals the IV (Wald) estimand when random assignment is viewed as the instrument.

3.5 A worked example

$N = 1{,}000$ subjects, 500 assigned $Z = 1$ and 500 $Z = 0$.

1. **Proportions.** In the $Z = 0$ group 100 take the drug: $\hat{\pi}_a = 100/500 = 0.20$. In the $Z = 1$ group 150 refuse: $\hat{\pi}_n = 150/500 = 0.30$. So $\hat{\pi}_c = 1 - 0.20 - 0.30 = 0.50$.
2. **ITT effects.** $\bar{Y}_{Z=1} = 7.0$, $\bar{Y}_{Z=0} = 6.0$, so $\mathrm{ITT}_y = 1.0$; $\bar{X}_{Z=1} = 0.70$, $\bar{X}_{Z=0} = 0.20$, so $\mathrm{ITT}_x = 0.50 = \hat{\pi}_c$.
3. **CACE** $= \mathrm{ITT}_y / \mathrm{ITT}_x = 1.0 / 0.50 = 2.0$.

The ITT (1.0) is attenuated by non-compliance, and the naive comparison of $X = 1$ with $X = 0$ would be confounded.

3.6 CACE estimator

For two-sided non-compliance,

$$\widehat{\text{CACE}} = \frac{\widehat{\text{ITT}}_y}{\hat{\pi}_c} = \frac{\sum_i Y_i Z_i / \sum_i Z_i - \sum_i Y_i(1-Z_i)/\sum_i (1-Z_i)}{1 - \sum_i X_i(1-Z_i)/\sum_i (1-Z_i) - \sum_i (1-X_i) Z_i / \sum_i Z_i},$$

with denominator $\widehat{\text{ITT}}_x = \hat{\pi}_c = 1 - \hat{\pi}_a - \hat{\pi}_n$. Standard errors via asymptotics or the bootstrap. Without A1–A3, nonparametric bounds can be obtained (Lecture 9) but are often too wide to be informative.

3.7 The "fourth condition": homogeneity

Under homogeneity the IV estimand identifies the ATE for the whole population. Three progressively weaker versions, any one sufficient:

(a) a constant effect for all units, $Y(1) - Y(0) = \beta_1$;

(b) equal average effects across Z, $\mathbb{E}[Y(1) - Y(0) \mid Z = 0] = \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$;

(c) no effect modification by U, $\mathbb{E}[Y(1) - Y(0) \mid U]$ constant.

(a) implies (b) implies (c).

(a) is rarely plausible; (b), (c) are weaker but still strong. Most applied work relies on monotonicity and the LATE interpretation, which needs only "no defiers" but identifies effects for compliers only (Hernán and Robins 2020, §§16.3–16.4).

3.8 Summary

- **Linear IV:** $\beta_1 = \text{Cov}(Y, Z)/\text{Cov}(X, Z)$; IV = ILS = 2SLS under just-identification; assumes a constant effect.
- **Potential-outcome IV / LATE:** under A1–A3, CACE $= \text{ITT}_y/\text{ITT}_x$, the average effect for compliers; heterogeneous effects allowed, at the price of a local estimand.
- Under constant effects, CACE = ATE $= \beta_1$.

A Candidate instruments in observational studies

Type	Example	Key concern
Genetic variant (Mendelian randomization)	ALDH2 polymorphism, alcohol metabolism	pleiotropy (direct effect on Y?)
Provider preference	physician's prescribing tendency	unmeasured patient factors drive referral
Geographic/temporal access	distance to facility, policy change date	correlated with SES, urban/rural

Only relevance is empirically testable; exclusion and no confounding require subject-matter knowledge. Always ask whether Z affects Y through any path other than X (*What If*, Fine Point 16.1).

B IV versus other causal methods

IV estimation	IPW / g-methods
Does not require conditional exchangeability $Y(x) \perp\!\!\!\perp X \mid W$; works with unmeasured confounding.	Requires conditional exchangeability (all confounders measured).
Sensitive to violations of exclusion/no confounding, with bias amplification.	Minor violations give minor bias (no amplification).
Identifies only a local effect (compliers) under monotonicity.	Identifies the population ATE; extends to time-varying treatments.

Prefer IV when unmeasured confounding is likely, a strong instrument exists, treatment is dichotomous, and either homogeneity or monotonicity is plausible (*What If*, §16.6).

C Practical issues with classic IV studies

Quarter of birth (Angrist and Krueger 1991): weak instrument (QOB explains $< 0.1\%$ of schooling variation); many-instruments bias (180 instruments push 2SLS toward OLS; Bound et al. 1995 obtained similar results with random data); exclusion (birth season may directly affect outcomes via health and school readiness); exogeneity (birth timing correlates with family background; Buckles and Hungerman 2013). Despite the critiques, influential in establishing the IV/LATE framework.
Proximity to college (Card 1995): exclusion (proximity correlates with urban/rural status, SES and school quality); monotonicity (living near a college may not uniformly increase schooling); LATE interpretation (the effect applies to compliers, those marginal with respect to proximity, likely lower-income, so LATE $\neq$ ATE) (Carneiro et al. 2011).

D Weak instruments

Three problems. (1) Wide confidence intervals: a weak Z–X association gives large variance of $\hat{\beta}_{1,\text{iv}}$.

(2) Bias amplification: $\hat{\beta}_1 = \text{ITT}_y/\text{ITT}_x$, so a small bias in the numerator from a minor exclusion violation is divided by a small denominator and becomes large.

(3) Finite-sample bias: even with a valid instrument, 2SLS is biased toward OLS in finite samples. Rule of thumb: first-stage $F < 10$ (the F-test of $H_0 : \alpha = 0$ in $X = \alpha_0 + \alpha Z + \epsilon$) indicates a weak instrument (Staiger and Stock 1997; *What If*, §16.5).

Example (*What If*, §16.2): effect of smoking cessation X on weight gain Y with cigarette price in the state of birth as instrument Z. $P(X = 1 \mid Z = 1) - P(X = 1 \mid Z = 0) = 25.8\% - 19.5\% = 6.3\%$ (weak), first-stage $F = 0.8 \ll 10$; IV estimate 2.4 kg with 95% CI $[-36.5, 41.3]$: uninformative. A weak instrument with even slight exclusion violations can produce arbitrarily large bias.

What to do. Report the first-stage F and compare it with the threshold for the number of instruments K (Stock and Yogo 2005). Use confidence sets valid under weak instruments: the Anderson–Rubin (AR) set inverts a test that bypasses the first stage, does not rely on instrument strength, and with a single instrument is optimal, so there is no reason not to report it. With multiple instruments prefer LIML (limited-information maximum likelihood) over 2SLS: it is less biased with many or weak instruments (Anderson and Rubin 1949; Andrews, Stock and Sun 2019; Montiel Olea and Pflueger 2013).

What not to do. Screening on $F > 10$ and reporting only passing specifications (pretesting worsens size distortion); trying many instrument sets and picking the highest F (instrument shopping, like p-hacking); bootstrapping IV inference (the bootstrap cannot handle the non-standard distribution under weak instruments); trusting conventional intervals $\hat{\beta} \pm 1.96\,\mathrm{SE}$ when F is small (normality fails). Weak instruments are a research-design problem, not just an estimation problem; the best solution is a stronger instrument.

E Many instruments and robust IV

With K instruments each gives $\hat{\beta}_k = \mathrm{Cov}(Y, Z_k)/\mathrm{Cov}(X, Z_k)$. If all are valid they agree on β_1; if some are invalid (e.g. pleiotropy) the estimates diverge, and 2SLS, which mixes them all, is contaminated. Robust alternatives: the weighted median (Bowden et al. 2016), consistent if more than 50% of instruments are valid; the mode-based estimator (Hartwig et al. 2017), consistent if a plurality agree; MR-Egger (Bowden et al. 2015), which allows directional pleiotropy through an intercept that corrects the bias. These are widely used in Mendelian randomization, where hundreds of genetic variants serve as instruments and some violate the exclusion restriction.

F Source coverage

The source deck has 53 physical pages (40 numbered slides).

Topic	Physical pages
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Linear IV: setup, ILS, 2SLS, covariates, derivations	7–19
Non-compliance: types, ITT, assumptions, identification, CACE, example	20–42
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